Technical Neural Networks Assignment Sheet 2b

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(Programming Assignment PA-A is solved separately.)

Assignment 7

The δ-rule as given in the lecture:

$$^{p}\Delta w_{gh} = \eta \cdot {}^{p}\delta_h \cdot {}^{p}\mathrm{out}_g$$

- $^{p}\Delta w_{gh}$: the change applied to the weight from neuron g to neuron h, for pattern p.
- η: the learning rate, controls the step size.
- $^{p}\delta_h$: the error signal of neuron h, defined as $\delta_h = -\partial E/\partial \mathrm{net}_h$. For an output neuron m: $\delta_m = (\hat{y}_m - y_m)\, f'(\mathrm{net}_m)$. For a hidden neuron h: $\delta_h = f'(\mathrm{net}_h) \sum_k \delta_k w_{hk}$, i.e. the δ's of the neurons it feeds into, propagated backward through the weights.
- $^{p}\mathrm{out}_g$: the output (activation) of the pre-synaptic neuron g, i.e. the value that travels along this connection.

Assignment 8

Weights are counted including the BIAS, so between a layer of N inputs and a layer of H neurons there are $(N+1)\cdot H$ weights.

Two hidden layers N-H_1-H_2-M:

$$W_2 = (N+1)H_1 + (H_1+1)H_2 + (H_2+1)M$$

One hidden layer N-H-M:

$$W_1 = (N+1)H + (H+1)M = H(N+1+M) + M$$

Setting $W_1 = W_2$ and cancelling M on both sides:

$$H(N+1+M) = (N+1)H_1 + H_1H_2 + H_2 + H_2M = (N+1)H_1 + H_2(H_1+1+M)$$

$$\boxed{H = \frac{(N+1)H_1 + H_2(H_1+1+M)}{N+1+M}}$$

For N-H_1-H_2-M = 5-21-30-4:

$$H = \frac{6\cdot 21 + 30\cdot(21+1+4)}{5+1+4} = \frac{126+780}{10} = \frac{906}{10} = 90.6$$

H must be an integer, so we round to $H = 91$. Check: original network has $W_2 = 6\cdot 21 + 22\cdot 30 + 31\cdot 4 = 126 + 660 + 124 = 910$ weights; the replacement 5-91-4 network has $W_1 = 6\cdot 91 + 92\cdot 4 = 546 + 368 = 914$ weights – very close to the original 910.

Assignment 9

Assume a simple Perceptron (inputs x_1, x_2, BIAS, weights w_1, w_2, w_b, step function) could realize XOR:

$$y = 1 \text{ if net} = w_1 x_1 + w_2 x_2 + w_b > 0, \qquad y = 0 \text{ otherwise.}$$

XOR requires $y(0,0) = 0,\ y(0,1) = 1,\ y(1,0) = 1,\ y(1,1) = 0$, which gives four inequalities:

$$(0,0):\ w_b \le 0 \quad \text{(I)} \qquad\qquad (0,1):\ w_2 + w_b > 0 \quad \text{(II)}$$

$$(1,0):\ w_1 + w_b > 0 \quad \text{(III)} \qquad\qquad (1,1):\ w_1 + w_2 + w_b \le 0 \quad \text{(IV)}$$

Adding (II) and (III):

$$w_1 + w_2 + 2w_b > 0 \implies w_1 + w_2 > -2w_b \quad \text{(V)}$$

From (IV): $w_1 + w_2 \le -w_b$. Combined with (V):

$$-2w_b < w_1 + w_2 \le -w_b \implies -2w_b < -w_b \implies w_b > 0$$

This contradicts (I), which requires $w_b \le 0$. No weights w_1, w_2, w_b can satisfy all four inequalities at once, so a simple Perceptron without a hidden layer cannot implement XOR. ■

Assignment 10b

$$^{p}E^{*}(w_{hm}) = \frac{1}{2} \sum_{m=0}^{M} (^{p}\hat{y}_m - {}^{p}y_m)^6$$

As in the lecture, we use gradient descent, $\Delta w_{hm} = -\eta\, \partial E^*/\partial w_{hm}$, and the chain rule through net_m:

$$\frac{\partial E^*}{\partial w_{hm}} = \frac{\partial E^*}{\partial y_m} \cdot \frac{\partial y_m}{\partial \text{net}_m} \cdot \frac{\partial \text{net}_m}{\partial w_{hm}}$$

Each factor:

$$\frac{\partial E^*}{\partial y_m} = \frac{1}{2} \cdot 6(\hat{y}_m - y_m)^5 \cdot (-1) = -3(\hat{y}_m - y_m)^5, \qquad \frac{\partial y_m}{\partial \text{net}_m} = f'(\text{net}_m), \qquad \frac{\partial \text{net}_m}{\partial w_{hm}} = \text{out}_h$$

So:

$$\frac{\partial E^*}{\partial w_{hm}} = -3(\hat{y}_m - y_m)^5\, f'(\text{net}_m)\, \text{out}_h$$

As in the lecture, define $\delta^*_m := -\partial E^*/\partial \text{net}_m$, so that the update keeps the usual $\Delta w_{hm} = \eta\, \delta^*_m\, \text{out}_h$ form:

$$\boxed{^{p}\delta^*_m = 3\,(^{p}\hat{y}_m - {}^{p}y_m)^5\, f'(^{p}\text{net}_m)}$$

$$^{p}\Delta w_{hm} = \eta\, {}^{p}\delta^*_m\, {}^{p}\text{out}_h = 3\eta\,(^{p}\hat{y}_m - {}^{p}y_m)^5\, f'(^{p}\text{net}_m)\, {}^{p}\text{out}_h$$

Assignment 11b

AI use: I used Claude (Anthropic) to work through this part, as the sheet requires. I gave it the assignment text directly and asked for the derivative calculations for tanh and the logistic function, shown step by step. The derivation worked on the first try and I checked each step by hand, so nothing was discarded.

Logistic function. $f(z) = (1 + e^{-z})^{-1}$. By the chain rule:

$$f'(z) = -(1 + e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

Since $1 - f(z) = 1 - \dfrac{1}{1 + e^{-z}} = \dfrac{e^{-z}}{1 + e^{-z}}$, we get

$$f(z)\,(1 - f(z)) = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \frac{e^{-z}}{(1 + e^{-z})^2} = f'(z)$$

$$\boxed{f'(z) = f(z)\,(1 - f(z))}$$

tanh. With $u = e^z - e^{-z}$, $v = e^z + e^{-z}$, so $\tanh(z) = u/v$, $u' = v$, $v' = u$. By the quotient rule:

$$\tanh'(z) = \frac{u'v - uv'}{v^2} = \frac{v^2 - u^2}{v^2} = 1 - \left(\frac{u}{v}\right)^2$$

$$\boxed{\tanh'(z) = 1 - \tanh^2(z)}$$

Assignment 12b

Claim: $\tanh(z) = 2f_{\log}(2z) - 1$, where $f_{\log}(z) = \dfrac{1}{1 + e^{-z}}$.

$$2f_{\log}(2z) - 1 = \frac{2}{1 + e^{-2z}} - 1 = \frac{2 - (1 + e^{-2z})}{1 + e^{-2z}} = \frac{1 - e^{-2z}}{1 + e^{-2z}}$$

Multiplying numerator and denominator by e^z:

$$\frac{1 - e^{-2z}}{1 + e^{-2z}} = \frac{e^z - e^{-z}}{e^z + e^{-z}} = \tanh(z)$$

So $\tanh(z) = 2f_{\log}(2z) - 1$: tanh is exactly the logistic function, stretched horizontally by a factor 2 and then shifted/rescaled from the range $(0, 1)$ to $(-1, 1)$. ■